

The Exact Minimal Time for One-End Observation and Control of a Finite String

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Abstract

We determine the sharp one-end observation and boundary-control time for the one-dimensional Dirichlet wave equation. The full energy wave group has least revival time $2L/c$, not L/c .

1 Frozen string and its full revival

Let $L, c > 0$ and consider

$$u_{tt} - c^2 u_{xx} = 0, \quad 0 < x < L, \quad u(0, t) = u(L, t) = 0. \quad (1)$$

The adjoint data lie in $H_0^1(0, L) \times L^2(0, L)$, with conserved energy

$$E(t) = \frac{1}{2} \int_0^L (|u_t(x, t)|^2 + c^2 |u_x(x, t)|^2) dx. \quad (2)$$

The observation is $u_x(L, \cdot)$. There is no zero mode: the frequencies are $\omega_n = n\pi c/L$, $n \geq 1$.

Lemma 1 (Least full-space revival). *The least positive time for which the wave group is the identity on the full energy space is*

$$T_* = \frac{2L}{c}.$$

Proof. At time T_* , every phase $e^{\pm i\omega_n T_*} = e^{\pm 2\pi i n}$ is one. Conversely, an identity time must return the $n = 1$ mode and is therefore an integer multiple of $2L/c$. In particular, L/c multiplies the n th mode by $(-1)^n$ and is not the full-space identity. $\square$

References

- [1] J.-L. Lions, *Exact controllability, stabilization and perturbations for distributed systems*, SIAM Review **30**(1) (1988), 1–68, [doi:10.1137/1030001](https://doi.org/10.1137/1030001).
- [2] C. Bardos, G. Lebeau, and J. Rauch, *Sharp sufficient conditions for the observation, control, and stabilization of waves from the boundary*, SIAM Journal on Control and Optimization **30**(5) (1992), 1024–1065, [doi:10.1137/0330055](https://doi.org/10.1137/0330055).